

The Spinor-Corrections
Theorem: exact quanta
 $bC = \pi^2/98$ and $aC = \pi^5/369754$

No κ -leaks in the geometry · diagonal holonomy or dead · phantom hits $\leq 3.10\%$ of $\kappa\pi/30$

Author:	Isahk Khamzatovich Isaev
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Status:	PROVED EXACTLY (S0–S4 layers, 12 equations)

Abstract

This supplementary monograph records the spinor-corrections theorem: the three repo corrections (bC, aC, aB) embedded into the corrected tower as natural coupling renormalizations are exact rational functions of π — the Berry quanta $bA = \pi^2/8$, $bB = \pi^2/18$, $bC = \pi^2/98$ and the braking quanta $aA = \pi^5/704$, $aB = \pi^5/5346$, $aC = \pi^5/369754$ — and the repo crosscheck closes at machine zero: $\Delta_{bC} - \lambda_{D^2} = bC$ (status OK), $\delta_{\text{eff}} = aC$ (absdiff 1.08×10^{-19}), $\gamma = \delta_{C^4}/22$ (absdiff 4.34×10^{-19}).

The geometry carries no κ -leaks: the SC/UV/Mdef doors are κ -free, and the C1-ring residues vanish only at $\kappa_{C1} = \kappa_{C2}$ — the holonomy correction must be diagonal. Of the twelve embedding schemes ten keep the CSS point alive; the phantom spectrum hits the spinor ladder $k\pi/30$ no closer than 2.60–3.10% in all three pencil schemes. Tables of the layers S0–S4, corollaries, honest caveats and the reproduction protocol are included.

Keywords: spinor corrections, $bC = \pi^2/98$, $aC = \pi^5/369754$, screening-off, diagonal holonomy, $k\pi/30$ ladder, SymPy, machine theorem.

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1. Introduction

This supplementary monograph of the theorems12 cycle records the spinor-corrections theorem — the machine result of the S0–S4 campaign (machine `sympy_spinor_corrections.py`, verdict `spinor_corrections.json`, LEDGER §22). The campaign embedded the three repo corrections bC (Berry), aC (braking) and aB into the towers as natural coupling renormalizations and asked whether they correct the towers and close on $\pi/30$.

The answer has three parts. First, the corrections themselves are exact: every quantum is a rational function of π , and the repo's own cross-identities ($\Delta_{bC} - \lambda_{D^2} = bC$, $\delta_{\text{eff}} = aC$, $\gamma = \delta_{C^4/22}$) close at 10^{-19} — the embedding is consistent with the stored constants. Second, the geometry admits the corrections only in diagonal form: the κ -free doors audit shows no leaks, and the C1-ring consistency forces $\kappa_{C1} = \kappa_{C2}$ (diagonal or dead). Third, at the linear level the corrections do not manufacture the spinor clock: the phantom spectrum approaches the $k\pi/30$ ladder only to 2.60–3.10%, so the screening-off verdict stands — the corrections renormalize, they do not select.

2. Notation and Setting

The three corrections act on the angles $\delta_A = \pi/2$, $\delta_B = \pi/3$, $\delta_C = \pi/7$ (the A/B/C spinor sectors). Each sector carries a Berry quantum b and a braking quantum a ; the repo corrections are the C-sector pair (bC, aC) together with the B-sector pair (bB, aB) used in the baryon-asymmetry reading. The corrected tower books acquire the factors $(1 \pm bC \mp aC)$ in the pencil schemes; the baseline scheme S0 runs without corrections.

Table 13.1. The exact spinor quanta (S0 layer)

Sector	Berry quantum b	Braking quantum a
A ($\delta_A = \pi/2$)	$\pi^2/8 = 1.2337005501$	$\pi^5/704 = 0.4346870523$
B ($\delta_B = \pi/3$)	$\pi^2/18 = 0.5483113556$	$\pi^5/5346 = 0.0572427394$
C ($\delta_C = \pi/7$)	$\pi^2/98 = 0.1007102490$	$\pi^5/369754 = 0.0008276305$

The setting: verify the repo cross-identities at machine zero (S0), audit the κ -doors of the twelve-equation chain (S1), test the embedding schemes (S2), and measure the phantom spectrum of the corrected pencils against the spinor ladder $k\pi/30$ (S4).

3. Statement of the Theorem

Theorem 13 (spinor corrections). Let the corrected tower be embedded with the repo corrections (bC, aC, aB). Then:

(i) the quanta are exact: $bA = \pi^2/8$, $bB = \pi^2/18$, $bC = \pi^2/98$, $aA = \pi^5/704$, $aB = \pi^5/5346$, $aC = \pi^5/369754$; the repo cross-identities $\Delta_{bC} - \lambda_{D^2} = bC$, $\delta_{\text{eff}} = aC$ and $\gamma = \delta_{C^4}/22$ hold to $\leq 4.34 \times 10^{-19}$ (S0 crosscheck status OK);

(ii) the geometry is κ -clean: the SC/UV/Mdef doors carry no κ ($\kappa_{\text{leaks_in_geometry}} = []$), the C1-ring residues factorize as $R1hT0h^2(K1 - K2)/2$, $-R1hT0h^2(K1 - K2)/3$ and $R1hT0h^2(K1 - K2)(K2T0h^2 + 7)/18$ and vanish exactly at $\kappa_{C1} = \kappa_{C2}$ — the holonomy correction is admissible only diagonally;

(iii) on the surviving diagonal branch the pair (UV_ ξ 3, Mdef_ ξ 5) reduces to branches $-2T0h^2(8\kappa^2T0h^2 - 93\kappa - 15KT)$ and $-2T0h^2(4\kappa^2T0h^2 - 39\kappa - 15KT)$ with resultant 0 and GCD $2T0h^2$; the uniform clock family of the embedding schemes is $\tau^*(\kappa) = 27/(2\kappa)$ exactly ($\text{uniform_family_is_27_over_2kappa} = \text{true}$);

(iv) of the twelve schemes ten keep the point alive and two are dead by construction (the family scheme and the violation demo); the phantom spectrum of the three pencil schemes hits the ladder $k\pi/30$ no closer than: baseline $\text{Im } \lambda = 0.10797 \approx \pi/30$ (3.10%), $H_{\text{mono}} (1 + bC - aC) \text{ Im } \lambda = 0.91564 \approx 9\pi/30$ (2.85%), $H_{\text{screen}} (1 - bC + aC) \text{ Im } \lambda = 0.10745 \approx \pi/30$ (2.60%) — the corrections do not manufacture the spinor clock at the linear level (screening-off).

4. Sketch of the Machine Proof

Layer S0 computes the quanta symbolically from the beta/alpha functions $\beta = d^2/2$, $\alpha = d^5/22$ ($k = b2(K3) = 22$) at the three sector angles and cross-checks the repo's stored identities in 50-digit arithmetic. Layer S1 substitutes the corrected chain forms into the twelve equations; the door map shows SC/UV/Mdef carry no κ , and the C1 group

factorizes by $(\kappa_{C1} - \kappa_{C2})$ — a polynomial computation in $Q(\kappa_{C1}, \kappa_{C2}, \kappa_{TH})$. Layer S2 solves the family conditions and finds the common root condition $15(\kappa - KT)/(8\kappa^2)$; ten schemes survive the point-survival test.

Layer S4 builds the quadratic eigenvalue problem of each pencil scheme at its clock point, filters genuine roots from phantoms by the relative $\sigma_{\min}$ test (machine zero $\approx 10^{-9}$ for genuine, $\geq 10^{-5}$ for phantoms), and measures the distance of the best phantom hit to the ladder $k\pi/30$. The machine runs 10.9 s (runtime_s); all layers are deterministic and offline.

Table 13.2. Layers of the verdict spinor_corrections.json

Layer	Statement	Status
S0 quanta	$bC = \pi^2/98$, $aC = \pi^5/369754$; repo identities close at 10^{-19}	exact + 50 digits
S1 doors	no κ -leaks; C1-ring vanishes iff $\kappa_{C1} = \kappa_{C2}$	exact (polynomial)
S2 schemes	10/12 schemes alive; uniform family $\tau^* = 27/(2\kappa)$	exact
S4 pencils	phantom hits to $k\pi/30$: 3.10% / 2.85% / 2.60%	numerical filter

5. Numerical Results

Table 13.3. The pencil schemes and their clock points (S4)

Scheme	τ^* at point	Best phantom $\text{Im } \lambda$	Gap to $k\pi/30$
S0 baseline (no corrections)	6.750	$0.10797 \approx 1 \cdot \pi/30$	3.10%
H_mono ($1 + bC -$ aC), screening off	6.137	$0.91564 \approx 9 \cdot \pi/30$	2.85%
H_screen ($1 - bC +$ aC)	—	$0.10745 \approx 1 \cdot \pi/30$	2.60%

The residual structure is stable across the schemes: 16 QEP roots split into 4 genuine ($\text{rel } \sigma_{\min} \sim 1.4 \times 10^{-9}$) and 12 phantoms ($\text{rel } \sigma_{\min} \geq 2 \times 10^{-4}$); the genuine block stays at $\lambda \approx 0$ (the $|\text{Im } \lambda| \leq 0.002$ band), and the phantom cloud keeps its distance from the spinor ladder in every scheme. The best H_{mono} hit at $9\pi/30$ moves the phantom, not the clock — exactly the screening-off separation recorded in the verdict lines.

6. Corollaries

Corollary 13.1 (the corrections are books, not dials). Since every quantum is an exact rational function of π and the repo identities close at 10^{-19} , the corrections are part of the algebraic bookkeeping of the tower — they renormalize the books but introduce no new free parameters.

Corollary 13.2 (diagonal or dead is the embedding law). The $C1$ -ring factorization (ii) upgrades the scheme audit of monograph No. 7 to an identity: any asymmetric embedding dies at the polynomial level, before any numerics. This is the falsifiable design rule for all future correction classes.

Corollary 13.3 (the spinor clock needs finite amplitude). The $\leq 3.10\%$ phantom gaps close the linear route to the spinor clock: together with the weight-rule and dd-closure theorems (monographs No. 4–5), the $k\pi/30$ structure can only be produced by finite-amplitude dynamics — the working separation on which the deep campaigns are built.

7. Honest Caveats

The phantom-gap numbers are produced by a threshold filter on 50-digit arithmetic: as in monograph No. 4, the filter is calibrated and its verdicts are filter-relative, though the genuine/phantom margin (10^{-9} versus 10^{-4}) is comfortable. The doors audit covers the twelve-equation chain in the T1c source form; other source forms may redistribute the doors, and the audit must be re-run. The uniform family $\tau^*(\kappa) = 27/(2\kappa)$ is proved for the embedding family tested; a structurally different embedding (nonlinear in κ) is outside the theorem. Finally, the H_{screen} clock point is recorded without a τ^* value in the verdict — the scheme keeps the point alive but the book was not separately computed; only the phantom census is quoted for it.

8. Reproducibility

Table 13.4. Reproduction protocol for Theorem 13

Step	Command	Result
1	python3 sympy_spinor_corrections.py	results/spinor_corrections.json (layers S0–S4)
2	python3 sympy_spectrum_tau_star.py	results/spectrum_tau_star.json (holonomy family crosscheck)
3	python3 -m pytest tests/ -q	58 passed (spinor-correction tests included)

The machine runs 10.9 s on an ordinary laptop (deterministic, single-threaded, offline).
The data correspond to commit 5b256cd; an autonomous copy lives in
einstein_direct/open9/machines/.

References

- [1] Repository wild8highlander/choptuik_ac_bc:
einstein_direct/sympy_spinor_corrections.py;
einstein_direct/results/spinor_corrections.json (campaign v11, LEDGER §22).
- [2] Ibid.: einstein_direct/sympy_spectrum_tau_star.py;
einstein_direct/results/spectrum_tau_star.json (the holonomy theorem).
- [3] Ibid.: einstein_direct/s4_berry_one_brick.py (the one-brick Berry screening).
- [4] Monographs No. 4, No. 5, No. 7 of the theorems12 cycle.